QT01 – Foundations of Quantum Mechanics
Portfolio (POF) – Week 8

Weight: 70% of module grade

Candidate number: 123456

Week 2

Problem 1 **[16 marks]**

At time $t = 0$ a particle is represented by the wave function

$$\psi(x,0) = \begin{cases} A\left(\dfrac{x}{a}\right), & 0 \le x \le a, \\ A\dfrac{(b-x)}{(b-a)}, & a \le x \le b, \\ 0, & \text{otherwise}, \end{cases}$$

where A, a, and b are positive real constants.

(a) Normalise ψ (that is, find A, in terms of a and b).

(b) Sketch $\psi(x,0)$, as a function of x.

(c) Where is the particle most likely to be found, at $t = 0$?

(d) What is the probability of finding the particle to the left of a? Check your result in the limiting cases $b = a$ and $b = 2a$.

(e) What is the expectation value of x?

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

State your final answer(s) clearly here.

Physical insight

Add a short physical interpretation here. Explain what the result means, why it matters, or what feature of the system it illustrates.

Problem 2 **[6 marks]**

In the concrete representation of bras and kets by row and column vectors, the inner product is defined in terms of components:

$$\langle B|A\rangle = \begin{pmatrix} \beta_1^* & \beta_2^* & \beta_3^* \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \beta_1^*\alpha_1 + \beta_2^*\alpha_2 + \beta_3^*\alpha_3.$$

The rule for inner products is essentially the same as for dot products: add the products of corresponding components of the vectors whose inner product is being calculated. Show that the inner product $\langle B|A\rangle$, defined as above, satisfies all the axioms of inner products (conjugate symmetry, linearity in the second argument, and positive-definiteness).

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

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Physical insight

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Week 3

Problem 1 **[12 marks]**

Show that the position operator

$$\hat{x} = x,$$

and momentum operator

$$\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x},$$

are Hermitian, in other words, that

$$\langle \phi | \hat{x}\psi \rangle = \langle \hat{x}\phi | \psi \rangle \qquad \text{and} \qquad \langle \phi | \hat{p}\psi \rangle = \langle \hat{p}\phi | \psi \rangle \,.$$

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

State your final answer(s) clearly here.

Physical insight

Add a short physical interpretation here. Explain what the result means, why it matters, or what feature of the system it illustrates.

Problem 2 **[8 marks]**

The wave function of an electron in the lowest energy state of a hydrogen atom is proportional to e^{-r/a_0}, with a_0 a constant and r the distance from the nucleus. Calculate the relative probabilities of finding the electron inside a region of volume $1.0\,\text{pm}^3$, located at

- the nucleus,
- a distance a_0 from the nucleus.

As the region of interest is so small, even on the scale of the atom, we can ignore the variation of ψ within it.

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

State your final answer(s) clearly here.

Physical insight

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Week 4

Problem 1 **[5 marks]**

Consider a system whose state is given in terms of an orthonormal set of three vectors: $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ as

$$|\psi\rangle = \frac{\sqrt{3}}{3}|\phi_1\rangle + \frac{2}{3}|\phi_2\rangle + \frac{\sqrt{2}}{3}|\phi_3\rangle\,.$$

(a) Verify that $|\psi\rangle$ is normalised. Then, calculate the probability of finding the system in any one of the states $|\phi_1\rangle$, $|\phi_2\rangle$ and $|\phi_3\rangle$. Verify that the total probability is equal to one.

(b) Consider now an ensemble of 810 identical systems, each one of them in the state ψ. If measurements are done on all of them, how many systems will be found in each of the states $|\phi_1\rangle$, $|\phi_2\rangle$ and $|\phi_3\rangle$?

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

State your final answer(s) clearly here.

Physical insight

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Problem 2 **[17 marks]**

(a) Show that the sum of two Hermitian operators is Hermitian.

(b) Suppose $\hat{Q}$ is Hermitian, and α is a complex number. Under what condition (on α) is $\alpha\hat{Q}$ Hermitian?

(c) When is the product of two Hermitian operators Hermitian?

(d) Show that the Hamiltonian operator

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)$$

is Hermitian.

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

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Physical insight

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Week 5

Problem 1 [20 marks]

Consider a system whose wave function at time $t = 0$ is given by

$$|\psi(x,0)\rangle = \frac{5}{\sqrt{50}}\,|\phi_0(x)\rangle + \frac{4}{\sqrt{50}}\,|\phi_1(x)\rangle + \frac{3}{\sqrt{50}}\,|\phi_2(x)\rangle\,,$$

where $|\phi_n(x)\rangle$ is the wave function of the nth excited state for a harmonic oscillator of energy

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right).$$

(a) Find the average energy of this system.

(b) Find the state $|\psi(x,t)\rangle$ at a later time and the average value of the energy; compare the result with the value obtained in (a).

Hint: the eigenvalue equation for the time-evolution operator is

$$\hat{U}\,|\phi_n(x)\rangle \equiv e^{-i\hat{H}t/\hbar}\,|\phi_n(x)\rangle = e^{-iE_nt/\hbar}\,|\phi_n(x)\rangle\,.$$

(c) Find the expectation value of the operator $\hat{x}$ with respect to the state $|\psi(x,t)\rangle$ (i.e., find $\langle\psi(x,t)|\hat{x}|\psi(x,t)\rangle$).

Hint: Use

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}^\dagger + \hat{a}\right).$$

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

State your final answer(s) clearly here.

Physical insight

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Problem 2 **[6 marks]**

Calculate the expectation value of the momentum for a harmonic oscillator prepared in a state which is a superposition of two number states,

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|\psi_1\rangle - i \, |\psi_2\rangle \right) .$$

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

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Physical insight

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Week 7

Problem 1 **[10 marks]**

Consider a two-spin composite state $|\psi\rangle$ prepared such that we can write

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow_A \downarrow_B\rangle - |\downarrow_A \uparrow_B\rangle \right),$$

where, for clarity, the subscripts A and B denote the subspace on which Alice and Bob will measure their spin components. First, Bob measures y-component of the spin with the operator τ_y, then Alice measures x-component with the operator σ_x.

(a) What is the expectation value of

$$\langle \sigma_x \tau_y \rangle = \langle \psi | \sigma_x \tau_y | \psi \rangle ?$$

(b) Calculate the expectation values of

$$\langle \tau_y \rangle = \langle \psi | \tau_y | \psi \rangle$$

and

$$\langle \sigma_x \rangle = \langle \psi | \sigma_x | \psi \rangle .$$

(c) What is the correlation between the two measurements

$$\langle \sigma_x \tau_y \rangle - \langle \sigma_x \rangle \langle \tau_y \rangle ?$$

Solution

Replace this placeholder with your own detailed derivation, making your algebra clear and showing the steps needed to reach the result.

Final answer

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Physical insight

Add a short physical interpretation here. Explain what the result means, why it matters, or what feature of the system it illustrates.